CSE 1141 - COMPUTER PROGRAMMING I
Programming Assignment # 2
DUE DATE: 30/10/2019 - 23:59 (No extension)

In this assignment, you will write a program that will calculate the price of different types of houses. Your program allows a real estate to enter customer's choices regarding a new house and from that information, itemize the price of the choices and finally determine the total price. The options and prices for the customer's choices are listed below.

Your program **will need** to accept responses with a **mix** of upper and lower case letters.

The House Option Rules:

1. There are four blocks and each block has different types of houses.
2. Houses in each block have different base prices for floor 1.
3. The following table presents the standard/optional/not available furniture for each type of house.

Block	Type	Base Price	Furnished	Storey Value	Max. Storey
A	1+1	45000	-	1.65 %	10
	2+1	65000	s	1.74 %	
B	1+1	50000	s	2.00 %	8
	2+1	72000	-	1.70 %	
	3+1	97500	o	1.75 %	
C	2+1	80000	o	1.60 %	10
	3+1	102400	s	1.85 %	
	4+1	137000	-	2.10 %	
D	3+1	119900	o	1.90 %	8
	4+1	165000	-	2.00 %	

- ‘s’ indicates that this house has already been furnished by default (with additional cost).
- ‘o’ indicates that this house might be furnished with an additional cost if the customer accepts.
- ‘-’ indicates that this house is not furnished for this type.

4. A and C are 10-storey blocks, B and D are 8-storey blocks.
5. The price of the house increases depending on house storey. For the first floor, your program will give the base price. For all other floors, your program will add an additional cost by calculating Base Price * Storey Value/100 * (Input Storey -1).
   Base Price and Storey Value are given in the above table and Input Storey is entered by the user.

   Example:

   Block A, 1+1 house on floor 1 price, 45000

   Block A, 1+1 house on floor 2 price, 45000 + 45000*(0.0165 * 1) = 45742.5

6. Furnished cost for each type of houses;

1+1	2+1	3+1	4+1
13000	19000	26000	38000

7. After retrieving the customer's choices, your program should add sales taxes to the total cost as the following:
    - Apply 1% KDV for the house with total costs less than 100,000.
    - Apply 8% KDV for the house with total costs higher than (or equal) 100,000.

Here, example runs for different cases are provided.

Example Run 1:

```
--- Buy a House ---
(A) Block A
(B) Block B
(C) Block C
(D) Block D
Which Block you prefer? (A, B, C, D): C
(2) 2+1
(3) 3+1
(4) 4+1
Which house type you prefer? (2, 3, 4): 2
Select storey? (max10): 1
Do you want furnitured house? (Y, N): N
Your house price without furnished:80000.0
1% KDV costs: 800.0
You select Block C, 2+1 type on floor 1 house with total cost: 80800.0
```

Example Run 2:

```
--- Buy a House ---
(A) Block A
(B) Block B
(C) Block C
(D) Block D
Which Block you prefer? (A, B, C, D): C
(2) 2+1
(3) 3+1
(4) 4+1
Which house type you prefer? (2, 3, 4): 2
Select storey? (max10): 5
Do you want furnitured house? (Y, N): Y
Your house price with furnished:104120.0
8% KDV costs: 8329.6
You select Block C, 2+1 type on floor 5 furnitured house with total cost: 112449.6
```

Example Run 3:

```
--- Buy a House ---
(A) Block A
(B) Block B
(C) Block C
(D) Block D
Which Block you prefer? (A, B, C, D): C
(2) 2+1
(3) 3+1
(4) 4+1
Which house type you prefer? (2, 3, 4): 3
Select storey? (max10): 2
Your house price with furnished:130294.4
8% KDV costs: 10423.552
You select Block C, 3+1 type on floor 2 furnitured house with total cost: 140717.952
```

Example Run 4:

```
--- Buy a House ---
(A) Block A
(B) Block B
(C) Block C
(D) Block D
Which Block you prefer? (A, B, C, D): C
(2) 2+1
(3) 3+1
(4) 4+1
Which house type you prefer? (2, 3, 4): 4
Select storey? (max10): 7
Your house price:154262.0
8% KDV costs: 12340.96
You select Block C, 4+1 type on floor 7 house with total cost: 166602.96
```

Note: Your program should execute correctly for different test cases.

Submission Instructions:

Please zip and submit all your files using filename YourNumberHW2.zip (ex: 150713852HW2.zip) to Canvas system (under Assignments tab). Your zip file should contain the followings:

1. Java source code (Pro2_150713852.java)
2. Java class file (Pro2_150713852.class)

Notes:

1. Write a comment at the beginning of each program to explain the purpose of the program. Write your name and student ID as a comment. Include necessary comments to explain your actions.
2. Select meaningful names for your variables.
3. You are allowed to use the materials that you have learned in lectures & labs.
4. Do not use the things that you did not learn in the course.
5. Each student should submit his/her own homework. You can discuss with your peers about the homework but you are not allowed to exchange code or pseudocode. This also applies to material found on the web. Should some submitted homework assignments be identical or suspected to be identical, all involved parties

will get a grade of **ZERO** from all homework. In case of any forms of cheating or copying, both giver and receiver are equally culpable and suffer equal penalties.
6. No late submission will be accepted.

Grading:
House Sales (88 points)

- Reading three input values from the user (12 points)

For Block A

- Calculate house price (5)
- KDV cost (5)
- Total house cost (6)

For Block B

- Reading an input value for furnished (4)
- Calculate house price (5)
- KDV cost (5)
- Total house cost (6)

For Block C

- Reading an input value for furnished (4)
- Calculate house price (5)
- KDV cost (5)
- Total house cost (6)

For Block D

- Reading an input value for furnished (4)
- Calculate house price (5)
- KDV cost (5)
- Total house cost (6)

Submission format (12 points)

- 1 java file + 1 class file
- Make sure that your class file can be executed on another computer.
- Make sure that the input/output of your program must be the same with the examples above (all informative strings & spaces).
- Comments are necessary!